JEI Journal of Electronic Imaging

RESEARCH PAPER

PiSOD: a plug-and-play pseudosupervised framework for semantic-structural alignment in salient object detection

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ABSTRACT. Salient object detection (SOD) simulates the human visual attention mechanism. This task focuses on automatically identifying the most visually salient regions or objects in an image. Although existing methods have achieved notable accuracy, they still rely on costly pixel-level annotations and lack sufficient semantic and structural constraints, particularly the effective integration of pixel-level details with object-level semantic and structural information, which often leads to blurred boundaries, semantic drift, and overfitting under limited samples or noisy labels. To address these issues, we propose a pseudosupervised and integrable SOD framework (PiSOD) that generates fused pseudolabels both semantically consistent and structurally complete, enabling the model to learn high-quality saliency representations under weak supervision. Furthermore, we introduce a Saliency Prior Alignment Module, seamlessly integrated into training, which provides boundary constraints via a structural prior branch, semantic guidance via a semantic prior branch, and a pseudolabel selection mechanism for effective cross-modal alignment, thus enhancing stability and boundary precision. The PiSOD framework supports two modes: fine-tuning existing SOD models to boost accuracy and robustness, and zero-shot inference that directly utilizes fused pseudolabels for efficient saliency prediction. Experimental results demonstrate that, compared with the baseline SOD models, our PiSOD-enhanced approach achieves notable improvements across multiple challenging datasets. Specifically, the integration of PiSOD results in an average mean absolute error (MAE) reduction of about 0.3% and an average improvement of 1.3% in the S metric across ECSSD, HKU-IS, DUT-O, PASCAL-S, and DUTS-TE. Moreover, these improvements are particularly evident in preserving structural integrity and semantic consistency, highlighting the effectiveness of our method in enhancing saliency detection without additional supervised training.

 [DOI: 10.1117/1.JEI.35.XX.XXXXXX]

Keywords: salient object detection; pseudosupervised learning; cross-modal fusion; semantic-structural alignment

Paper 251565G received Nov. 3, 2025; revised Dec. 30, 2025; accepted Jan. 16, 2026.

1 Introduction

Salient object detection (SOD) aims to simulate the human visual attention mechanism by 1
automatically identifying the most visually distinctive regions or objects in an image.[1–3] As a fundamental task in computer vision, SOD supports a wide range of important downstream

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1017-9909/2026/$28.00

applications such as object recognition, image segmentation, visual tracking, and image captioning,[4,5] providing a crucial foundation for many intelligent visual systems. With the rapid development of deep learning, particularly convolutional neural networks (CNNs) and Transformer architectures,[6,7] SOD models have achieved remarkable improvements in both accuracy and generalization, enabling their application in complex real-world scenarios such as autonomous driving, medical image analysis, and human–computer interaction.[8]

However, as shown in Fig. 1(a), existing SOD methods still suffer from inherent limitations. They heavily rely on large-scale pixel-level annotations,[4] which are costly to obtain and difficult to extend to unlabeled or weakly labeled scenarios.[9] Moreover, most existing approaches primarily focus on local saliency cues while neglecting global semantic consistency and object structure,[10] leading to issues such as semantic drift, blurred boundaries, and poor robustness in complex scenes. This limitation is evident in existing pseudosupervised SOD methods, which often emphasize coarse pseudolabel generation or noise suppression but lack joint modeling of object-level semantic consistency and structural integrity. In addition, insufficient structural regularization during training often causes severe overfitting when data are scarce or noisy,[11] reducing model stability and generalization. In summary, current SOD methods are mainly constrained by their heavy dependence on dense pixel-level supervision, insufficient integration of fine-grained pixel-level details with object-level semantic-structural information, and limited robustness under weak or noisy supervision. These inherent limitations strongly motivate the need for a more flexible and structurally aware learning paradigm.

To overcome these challenges, we propose PiSOD, a pseudosupervised and integrable SOD framework that effectively achieves semantic-structural alignment by leveraging complementary cross-modal priors, as illustrated in Fig. 1(b). Unlike prior pseudosupervised methods that treat semantic cues and structural constraints separately or implicitly, PiSOD formulates pseudosupervised saliency detection as a unified semantic-structural alignment problem, enabling coherent and structurally consistent saliency learning. The core of PiSOD is the saliency prior alignment module (SPAM), a flexible plug-and-play training-stage component designed to jointly model semantic and structural information. Specifically, SPAM employs a structural prior branch to provide boundary-level constraints and a semantic prior branch to enhance contextual understanding, thereby enabling the generation of high-quality pseudolabels under pseudosupervision.[12]

Built upon this module, PiSOD supports two complementary operation modes. In the Fine-tuning Mode, these pseudolabels serve as supervision to refine existing SOD models, further improving their robustness and generalization.[9] In the Zero-shot Inference Mode, the saliency map generated by SPAM can be directly used as the final prediction without requiring pixel-level annotations.[13,14] Through this unified design, PiSOD effectively bridges semantic and structural priors, offering an efficient, annotation-free, and adaptable solution for salient object detection.

The main contributions of this work are summarized as follows:

- We propose the PiSOD framework, which explicitly models pseudosupervised saliency detection as a semantic-structural alignment problem and leverages pseudolabel generation with cross-modal selection to produce semantically consistent and structurally complete saliency predictions without manual annotations.

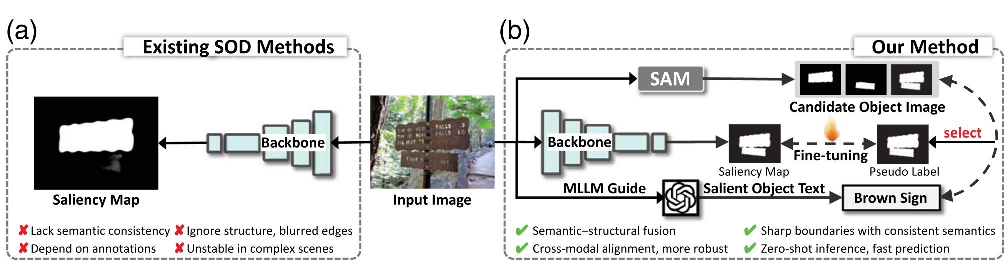

Fig. 1 Comparison between existing SOD methods and the proposed PiSOD framework. (a) Existing methods depend on dense annotations and often yield inconsistent semantics and blurred boundaries. (b) The proposed PiSOD framework adopts a pseudosupervised scheme that aligns structural and semantic priors to produce sharp, consistent saliency maps. It supports both fine-tuning and zero-shot inference for efficient, annotation-free detection.

- We design an integrable SPAM, a plug-and-play, training-stage component decoupled from backbone architectures, which enhances boundary precision, semantic consistency, and robustness without modifying existing SOD networks.
- We conduct extensive experiments on multiple public datasets and show that our proposed method outperforms state-of-the-art approaches in structural completeness and semantic consistency while also enabling zero-shot saliency inference.

2 Related Work

2.1 Fully Supervised SOD Methods

In SOD, fully supervised methods based on fine-grained pixel-level annotations have long dominated the field. These approaches typically leverage the powerful feature extraction capabilities of deep CNNs,[6,15,16] significantly enhancing saliency perception through multilevel feature fusion, as demonstrated in works by Chen et al.,[16] Zhang et al.,[17] and Pang et al.[18] To further improve the integrity and sharpness of object boundaries, some studies (e.g., Qin et al.,[11] Wang et al.,[19] Zhao et al.[20]) have introduced edge-guided supervision mechanisms. Representative models such as U-Net,[21] DSS,[15] PiCANet,[22] PoolNet,[10] and BASNet[23] have achieved state-of-the-art performance on public benchmarks such as DUTS,[4] ECSSD,[24] and PASCAL-S[25] by employing cross-layer connections, attention mechanisms, and deep supervision strategies.[8,10] Other studies have also explored the use of superpixels, multiscale context, and global recurrent structures to enhance model discriminability.[26,27] However, these methods heavily rely on high-quality pixel-level annotations, which are costly and time-consuming to obtain.[2] Such annotations require extensive manual effort and are often affected by inconsistencies and subjectivity, limiting dataset scalability and hindering generalization. Moreover, supervision in these methods is predominantly imposed at the pixel level, with limited explicit modeling of object-level semantic consistency and structural integrity, which may often lead to less reliable and unstable predictions in real-world applications when annotations are sparse or noisy. Although some research has explored semisupervised or weakly supervised mechanisms to reduce dependency on annotations,[9] there remains a notable performance and stability gap compared with fully supervised methods. Therefore, achieving high-quality saliency detection under weakly supervised or unsupervised conditions has become a key research focus and challenge,[5] motivating this paper's exploration of pseudosupervised approaches for SOD.

2.2 Weakly and Pseudosupervised SOD Methods

To reduce reliance on pixel-level annotations, SOD has recently expanded toward weakly supervised and pseudosupervised learning paradigms. Weakly supervised methods typically use low-cost supervisory signals such as image-level labels, category tags, image captions, or scribble annotations. For example, Wang et al.[4] proposed a foreground inference network combined with global smooth pooling to mine salient regions from image-level classification labels.[9] Zhang et al. introduced a scribble annotation approach that allows each image to be labeled in just 1 to 2 s, enabling the construction of a scribble-based dataset and related learning strategies.[9] Yu et al.[28] further incorporated self-consistency mechanisms and feature aggregation modules to enhance model generalization. Zeng et al.[29] integrated image captions and category labels and designed an attention transfer mechanism to improve weak-label learning. Although these methods significantly reduce annotation costs, issues such as pseudolabel noise, semantic drift, and limited generalization remain unresolved. In particular, most existing approaches rely heavily on local pixel-level saliency cues or heuristic pseudolabel generation while lacking explicit modeling of object-level semantic consistency across the entire scene. Consequently, saliency predictions within the same object tend to be inconsistent, and semantic confusion is likely to occur in complex scenes, especially those with similar foreground–background appearances or multiple salient objects. To eliminate manual annotation entirely, some studies have explored unsupervised saliency detection. These approaches often use heuristic-based methods (e.g., RBD, MC, HS) to generate initial labels, then refined using fusion strategies,[25,30] label enhancement,[17] or noise modeling.[26] Although these approaches avoid manual involvement, the quality of pseudolabels heavily depends on handcrafted heuristics, limiting their robustness and transferability. Moreover, most of these methods do not explicitly incorporate object-level structural priors or global shape constraints, often leading to fragmented predictions, blurred boundaries, or incomplete representations of thin, elongated structures. These issues are further amplified under weak supervision or noisy pseudolabel conditions, where the lack of effective structural regularization can cause unstable training and overfitting.

To address the limitations of prior methods in label quality, structure modeling, and semantic consistency, we propose PiSOD, a pseudosupervised saliency detection framework without annotations. PiSOD employs a structure-semantic dual-guided mechanism and integrates multimodal information from images and text to generate high-quality pseudolabels. It supports saliency prediction without training, greatly improving applicability in label-free and cross-domain scenarios.

2.3 Robust and Modular Optimization Strategies

In the study of SOD, challenges such as model overfitting, limited generalization, and unstable feature representation have received considerable attention.[17] Particularly under conditions of scarce annotated data or high label noise, saliency models often learn redundant or nondiscriminative features, leading to unstable training and degraded performance during testing. This instability is closely related to the absence of explicit object-level semantic alignment and structural regularization, which makes models more prone to overfitting pixel-level noise and local artifacts, rather than effectively capturing coherent, semantically meaningful object representations across the entire scene. To alleviate these issues, recent studies have incorporated various strategies, including structural regularization,[31] boundary refinement,[23] attention mechanisms, and multitask learning,[20,22] to improve model robustness and feature consistency. Approaches such as boundary-aware modules, deep supervision, and contrastive learning[32] have shown notable gains in performance. However, most of these strategies are tightly coupled with specific network architectures, lacking universality, modularity, and extensibility across different frameworks.

To bridge this gap, we propose the SPAM, a general and integrable component designed to enhance structural regularization and feature consistency in SOD networks. SPAM introduces an alignment mechanism that optimizes the distribution of saliency features by jointly considering semantic coherence and boundary precision, thereby mitigating overfitting and improving overall robustness during training. With its strong structural generality and flexible plug-and-play integration, SPAM can be seamlessly embedded into a wide range of existing SOD architectures without modifying their backbones. This design not only strengthens the stability and generalization of saliency detection models but also offers a unified and extensible framework for future research in this domain.

3 Method

3.1 Overview

Although existing SOD methods have made progress in localizing salient regions, most approaches still rely heavily on pixel-level annotations[6,15] and suffer from limitations in semantic consistency, boundary completeness, and cross-modal adaptability.[33] To address these issues, we propose PiSOD, a unified pseudosupervised SOD framework that supports both fine-tuning and zero-shot inference. In the fine-tuning mode, PiSOD introduces a SPAM to generate high-quality pseudolabels by jointly exploiting object-level structural priors and cross-modal semantic guidance, enabling backbone-agnostic refinement of existing SOD models with improved semantic consistency and boundary integrity.[22] In the zero-shot mode, PiSOD directly produces saliency maps from fused SAM masks[14] without additional training or manual annotations, enabling fast and lightweight inference. Overall, PiSOD achieves effective semantic-structural alignment while maintaining flexibility and compatibility across diverse backbone architectures. Notably, SPAM is a backbone-agnostic, plug-and-play module that can be seamlessly integrated into CNN-based, Transformer-based, or hybrid-based SOD networks without architectural modifications, ensuring broad compatibility while preserving semantic consistency and structural integrity. Compared with existing pseudosupervised methods, PiSOD explicitly aligns semantic and structural cues to generate high-quality pseudolabels, leading to improved intra-object coherence and boundary completeness under weak supervision.

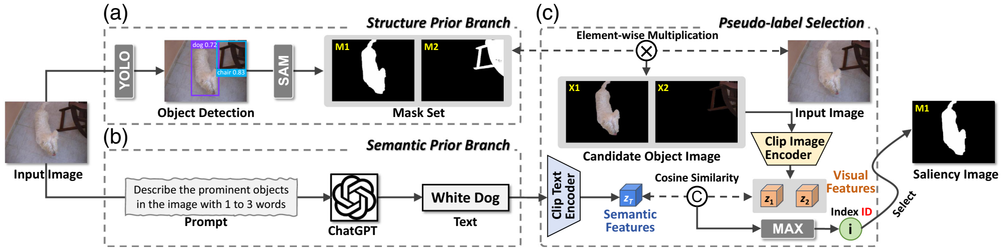

Fig. 2 Overall architecture of the proposed saliency prior alignment module (SPAM). (a) The structure prior branch extracts object-level masks using YOLO and SAM. (b) The semantic prior branch generates text descriptions of salient objects via a multimodal language model. (c) The pseudo-label selection aligns semantic and structural cues in the CLIP space to select the most relevant mask and produce the final saliency map.

As illustrated in Fig. 2, the SPAM module serves as a plug-and-play training component that can be seamlessly integrated into any SOD network. By leveraging semantic alignment and structural prior constraints, it provides cross-modal saliency supervision, enhancing the network's perception of salient regions and boundary delineation. Based on this design, the PiSOD framework not only supports fine-tuning existing SOD models to improve accuracy and robustness during training but also enables zero-shot saliency prediction by directly using SAM masks at inference.

In summary, SPAM focuses on training-stage saliency enhancement through structural and semantic alignment, whereas the PiSOD framework unifies fine-tuning and zero-shot inference under a single, coherent methodology. This unified design enables consistent performance improvement in annotated settings and rapid deployment in annotation-free scenarios, forming a robust, extensible, and practical solution for comprehensive saliency detection.

3.2 Saliency Prior Alignment Module

To address the limitations of SOD in semantic consistency and structural integrity, we design SPAM. During training, SPAM introduces semantic and structural constraints as regularization. It can be combined with any existing SOD network without modifying the backbone, significantly improving prediction stability and boundary quality.

3.2.1 *Structure prior branch*

Conventional SOD methods rely on pixel-level supervision but often lack object-level structural modeling, leading to blurred boundaries or broken predictions in complex scenarios.[20,23] Previous studies have shown that object detectors provide stable object localization for visual tasks,[34,35] whereas prompt-driven segmentation models can produce fine-grained object boundaries.[14] Inspired by this, SPAM incorporates an object-level detection (e.g., YOLO) and prompt-driven segmentation collaboration (e.g., SAM) to generate reliable structural priors.

Given an input image I, we first apply an object detector to localize potential regions of interest, obtaining a set of candidate bounding boxes $\{B_i\}_{i=1}^{K}$ that roughly indicate the positions of salient or meaningful objects in the scene

$$\{B_i\}_{i=1}^{K} = \mathcal{D}(I), \tag{1}$$

where $\mathcal{D}(\cdot)$ denotes the detector and K is the number of detected candidates. Each candidate box B_i is then fed as a prompt to the prompt-driven universal segmentation model to obtain object-level masks M_i

$$M_i = \mathcal{S}(I, B_i), \qquad i = 1, \dots, K, \tag{2}$$

where $\mathcal{S}(\cdot)$ denotes the segmenter. The resulting mask set $M_{ii=1}^{K}$ covers the salient objects in the image and provides a reliable prior for subsequent structural constraints.

This branch combines the global position awareness provided by the object detector with the fine boundaries generated by the universal segmentation model to supply object-level structural priors for salient object detection. Such priors alleviate boundary blurring and target fragmentation in complex scenes while providing stable base regions for the semantic prior branch and pseudolabel selection, contributing to joint improvements in semantic consistency and structural integrity.

3.2.2 *Semantic prior branch*

In salient object detection, relying solely on pixel-level supervision can provide precise spatial constraints but often lacks global semantic and structural priors, which may result in background regions being misclassified as salient objects or a single object being predicted as disconnected fragments. To enhance the model's capability in capturing high-level semantics and object boundaries, we introduce a semantic prior branch within the Saliency Prior Alignment Module to provide additional semantic constraints and improve prediction consistency and completeness.

Specifically, we use a multimodal large language model (MLLM, e.g., GPT) using a prompt P to generate a concise semantic description T of the most salient object from the input image, such as "dog" or "person."[36–38] This process can be formulated as

$$T = \mathrm{MLLM}(I, P), \tag{3}$$

where I denotes the input image and P is the designed prompt instructing the model to describe only the most salient object in the image and limit the output to one to three words (a noun or noun phrase), thus avoiding long or scene-based descriptions. The final semantic prior is denoted as T.

Unlike traditional pixel-based annotation approaches, this process requires no extra human labeling; instead, it automatically obtains high-level semantic information through the cross-modal reasoning capability of large models. During training, the semantic prior serves as a semantic anchor that guides the model's saliency prediction to better correspond to the most attention-grabbing object in the image. In addition to contributing to the semantic loss, this branch also provides a global semantic reference for the structure prior branch, helping to reduce semantic drift and making saliency prediction more consistent at the object level.

3.2.3 *Pseudolabel selection*

Although the structure prior branch can provide multiple candidate object masks for the input image, not all correspond to the most salient object, and directly using them may introduce noise. Therefore, we design a selection module that automatically identifies objects most consistent with saliency semantics by combining the semantic prior with a cross-modal alignment mechanism.

Specifically, given the candidate mask set $\{M_i\}_{i=1}^{N}$ from the structure prior branch, we perform elementwise multiplication between each mask and the original image to obtain candidate object images X_i

$$X_i = I \odot M_i, \tag{4}$$

where I denotes the input image, M_i denotes the i'th candidate mask, and $\odot$ represents elementwise multiplication. The resulting X_i thus retains only the pixel information within the corresponding mask region for subsequent processing.

Next, we use CLIP's image encoder to extract visual features for each candidate object $\mathbf{z}_i = f_v(X_i)$.[36,39] Meanwhile, the semantic prior branch generates the salient object description T based on GPT,[40] and CLIP's text encoder extracts the corresponding semantic feature $\mathbf{z}_T = f_t(T)$. We then compute the cosine similarity s_i between the semantic and visual features

$$s_i = \cos(\mathbf{z}_i, \mathbf{z}_T), \tag{5}$$

where $\cos(\mathbf{z}_i, \mathbf{z}_T) = \frac{\mathbf{z}_i \cdot \mathbf{z}_T}{\|\mathbf{z}_i\| \|\mathbf{z}_T\|}$ is used to measure the alignment between the candidate image and the semantic description; the larger the value, the better the match.

We assess the match degree between each candidate mask and the semantic prior, sort the similarities s_i in ascending order, find the maximum adjacent difference to determine a threshold τ, and retain masks with $s_i \geq \tau$. These masks are then fused into the final mask M_{fused}

$$M_{\mathrm{fused}} = \bigcup_{i : s_i \geq \tau} M_i, \tag{6}$$

where τ is the threshold determined by the maximum adjacent difference and $\bigcup$ denotes the pixel-level union operation. All masks meeting the similarity condition are merged to obtain the final mask M_{fused}, ensuring consistency with both the structural and semantic priors.

The core of this module lies in integrating high-level semantic priors with object-level structural priors through cross-modal alignment to automatically select candidate masks. The fused mask M_{fused} exhibits high boundary completeness and semantic consistency with the most salient object in the image (Sec 3.3). It can serve as input for fine-tuning SOD models and also act as the direct output for saliency prediction, providing a reliable supervisory signal for subsequent optimization.

3.3 Training and Inference

3.3.1 *Loss functions*

Semantic loss. In this study, the semantic loss is designed as a negative logarithm of cosine similarity computed on the fused image. As illustrated in Fig. 3, the purpose is to leverage the fused image to highlight salient regions and suppress background interference, making the extracted visual features more focused on the foreground object. The negative logarithmic form imposes stronger penalties on semantically inconsistent samples, which improves the precision of semantic supervision and ensures numerical stability without extra hyperparameter tuning.[6,32]

We use an existing salient object detection network to predict the pixel-level saliency map $\hat{M}$ of the input image I

$$\hat{M} = \mathcal{F}_{\text{SOD}}(I), \tag{7}$$

where $\mathcal{F}_{\text{SOD}}(\cdot)$ denotes the salient object detection model and $\hat{M} \in [0,1]^{H\times W}$ is the predicted saliency probability map. Compared with the prompt-based SAM segmentation results, the saliency map can provide a globally consistent saliency distribution, which is used together with object-level masks to construct structural constraints.

To introduce accurate cross-modal semantic constraints in the saliency detection task, we first perform elementwise multiplication between the input image I and the predicted saliency map $\hat{M}$ to obtain a fused image I_{fused}

$$I_{\text{fused}} = I \odot \hat{M}, \tag{8}$$

where I denotes the input image, $\hat{M}$ denotes the predicted saliency map, and $\odot$ denotes elementwise multiplication. This process corresponds to the fusion operation shown in Fig. 3, where salient regions are emphasized for subsequent semantic encoding.

The fused image I_{fused} is then fed into the CLIP image encoder to obtain visual features $v_I = f_v(I_{\text{fused}})$, and the text label T is fed into the CLIP text encoder to obtain text features

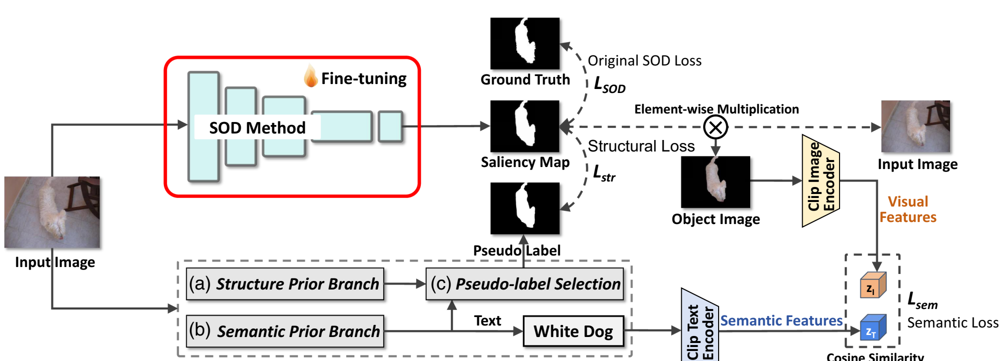

Fig. 3 Overview of the fine-tuning stage of the proposed PiSOD framework. The SPAM module produces pseudolabels via (a) structure prior branch, (b) semantic prior branch, and (c) pseudo-label selection. These pseudolabels guide the fine-tuning of SOD models with structural loss ($\mathcal{L}str$) and semantic loss ($\mathcal{L}sem$) to improve boundary sharpness and semantic consistency.

$v_T = f_t(T)$. To enhance the penalty on semantically inconsistent samples, we take the negative logarithm of the normalized cosine similarity and define the semantic loss $\mathcal{L}_{\text{sem}}$ as

$$\mathcal{L}_{\text{sem}} = -\log\left(\frac{1+\cos(v_I, v_T)}{2} + \epsilon\right), \tag{9}$$

where ϵ is a small constant to ensure numerical stability.

Note: Using $\frac{1+\cos(v_I,v_T)}{2}$ instead of $1-\cos(v_I, v_T)$ ensures that the logarithm input remains non-negative and within [0,1], which prevents numerical instability and gradient explosion. It also produces larger gradients when similarity is low, which is beneficial for training.

Structure loss. In salient object detection, relying solely on pixel-level errors [e.g., mean squared error (MSE)] often cannot ensure the shape integrity and boundary precision of the predicted mask. As shown in Fig. 3, we therefore propose a joint structural loss to constrain both the predicted mask $\hat{M}$ from the SOD network and the fused SAM mask M_{fused} (see Sec 3.2.3) in terms of pixel and texture features. This ensures shape accuracy at the global pixel level and consistency in local boundaries and details. Specifically, the pixel-level error enforces regional alignment, whereas the texture structure constraint emphasizes boundaries and fine details, enabling stable and precise structural supervision without extra hyperparameter tuning and improving the structural integrity and boundary clarity of predicted masks.

This structural loss consists of two components:

A. *Pixel-level error constraint.* We use MSE to measure the consistency between the predicted mask and the fused SAM mask at the pixel level, and define this loss as $\mathcal{L}_p$

$$\mathcal{L}_p = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} (\hat{M}_{i,j} - M_{\text{fused},i,j})^2, \tag{10}$$

where H and W are the height and width of the mask, respectively. This loss ensures pixel-level alignment between the predicted mask and the target mask.

B. *Texture structure constraint.* To capture local texture features and boundary information of salient objects, we use the Laplacian operator $\mathcal{L}ap(\cdot)$ to extract texture responses from the predicted mask and the fused SAM mask, and then compute the texture differences $\mathcal{L}_t$

$$\mathcal{L}_t = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} |\mathcal{L}ap(\hat{M})_{i,j} - \mathcal{L}ap(M_{\text{fused}})_{i,j}|, \tag{11}$$

where $\mathcal{L}ap(\cdot)$ denotes the Laplacian filtering operation, used to extract local texture and boundary information to ensure precision of the predicted mask in local structures.

Finally, the joint structural loss $\mathcal{L}_{\text{str}}$ is defined as the sum of the pixel error $\mathcal{L}_p$ and texture structure constraints $\mathcal{L}_t$

$$\mathcal{L}_{\text{str}} = \mathcal{L}_{\text{p}} + \mathcal{L}_{\text{t}}. \tag{12}$$

Both components correspond to the structural consistency path in Fig. 3, where the predicted saliency map and pseudolabel jointly contribute to structural supervision. In practice, both losses are normalized and directly added without manually setting weights, forming a joint structural loss that ensures pixel accuracy while enhancing overall structural consistency and boundary clarity of salient objects, providing stable and reliable constraints for the salient object detection task.

3.3.2 *Fine-tuning mode*

As illustrated in Fig. 3, the fine-tuning mode integrates the semantic and structural constraints into existing SOD models. Unlike traditional fully supervised SOD training that relies solely on manual pixel-level annotations, PiSOD fine-tuning leverages fused pseudolabels generated by SPAM, providing cross-modal guidance to enhance both semantic consistency and boundary precision. To further improve salient object detection performance, we fine-tune the existing pretrained SOD model by jointly optimizing the proposed semantic loss $\mathcal{L}_{\text{sem}}$ and structural loss

$\mathcal{L}_{\text{str}}$ together with the original SOD loss $\mathcal{L}_{\text{SOD}}$. During training, the input image I is processed by the SOD model $\mathcal{F}_{\text{SOD}}$ to produce the predicted saliency map $\hat{M}$, which is used to compute the total loss $\mathcal{L}_{\text{total}}$ together with the fused mask M_{fused} obtained from the pseudolabel filtering mechanism in Sec 3.2 and the text label T

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{SOD}} + \mathcal{L}_{\text{sem}} + \mathcal{L}_{\text{str}}, \tag{13}$$

where $\mathcal{L}_{\text{SOD}}$ denotes the original supervisory loss of the saliency detection network (e.g., binary cross-entropy, IoU, or Dice loss), ensuring the basic pixel-level accuracy of the predicted mask; $\mathcal{L}_{\text{sem}}$ constrains the semantic alignment of the predicted map through cross-modal cosine similarity between the fused image and the text label; and $\mathcal{L}_{\text{str}}$ enhances the global structural consistency and boundary precision of the predicted mask via pixel-level error and texture-structure constraints.

During training, the CLIP encoders are frozen and only the SOD network weights are optimized end-to-end. This allows the model to retain its original saliency detection ability while effectively learning cross-modal semantic and structural priors, as shown in Fig. 3. Consequently, the fine-tuned model achieves more accurate localization, sharper boundaries, and improved semantic coherence under limited-annotation or weakly supervised conditions, distinguishing PiSOD fine-tuning from conventional fully supervised SOD training.

3.3.3 *Zero-shot inference mode*

In the zero-shot inference mode, PiSOD can directly generate saliency maps without pixel-level annotations or training. Specifically, the input image undergoes SAM segmentation and the pseudolabel filtering mechanism described in Sec 3.2 to obtain the fused mask M_{fused}, which is directly used as the predicted saliency map $\hat{M}$

$$\hat{M} = M_{\text{fused}}. \tag{14}$$

This approach requires no loss computation and does not rely on any pretrained SOD model, providing a fast estimate of foreground regions. Although its saliency prediction accuracy is generally lower than fine-tuned or fully trained SOD models, it is well suited for unlabeled data or rapid prototyping, demonstrating PiSOD's practicality and convenience under weak supervision.

4 Experiment

4.1 Datasets and Evaluation Metrics

We qualitatively and quantitatively compare different methods and their performance on five commonly used benchmark datasets: ECSSD,[24] HKU-IS,[41] DUT-O,[30] PASCAL-S,[25] and DUTS-TE.[4]

ECSSD contains 1000 images with a significant amount of foreground-background interference. HKU-IS includes 4447 images with multiple disconnected salient objects. DUT-O consists of 5168 images with complex structures and challenges in salient object detection. PASCAL-S comprises 850 images extracted from PASCAL VOC 2010, with additional salient object segmentation annotations. DUTS-TE is composed of 10,553 training images selected from ImageNet and 5019 test images selected from ImageNet and the SUN dataset. These data types are very diverse, including animals, people, plants, vehicles, aircraft, buildings, and household items, covering almost all categories and scenes required for detection. In addition, these datasets contain many more complicated scenes, including appearance change, big objects, clutter, shape complexity, and small objects.

To comprehensively evaluate the performance of the proposed method and compare it with other state-of-the-art approaches, we adopt four widely used metrics in salient object detection: MAE,[42] maximum F-measure (maxF),[43] enhanced-alignment measure (E-measure),[44] and structure-measure (S-measure).[45] These metrics jointly assess pixel-level accuracy, structural consistency, and perceptual alignment between the predicted saliency map and ground truth. MAE measures the average pixel-wise difference between prediction and ground truth, where lower values indicate higher overall similarity. The F-measure combines precision and recall to balance detection accuracy and completeness, with maxF representing the maximum obtained across different thresholds. The E-measure evaluates both local pixel-level correspondence and global perceptual alignment, providing a robust assessment of visual consistency. Finally, the S-measure assesses structural similarity by integrating region-aware and object-aware components, reflecting how well the predicted saliency map preserves object structures and boundaries.

4.2 Implementation Details

Our work is implemented in PyTorch and evaluated on an NVIDIA RTX 3090 GPU. Standard data augmentation techniques, including random flipping, rotation, scaling, and color jittering, are applied during training to enhance generalization.[46] For fine-tuning experiments, pre-trained salient object detection networks such as TRACER,[47] InSPyReNet,[48] M3Net,[49] MENet,[50] BBRF,[51] ADMNet,[52] 3SD,[53] DiffSOD,[54] DC-Net,[55] and UniSOD[56] are adopted as backbones. The SPAM module generates fused pseudolabels, and only the SOD network weights are optimized end-to-end while CLIP parameters remain fixed. The total loss combines the original SOD loss with the proposed semantic and structural losses. During inference, the framework supports two modes: fine-tuning mode, which uses both SOD network predictions and fused masks for saliency estimation, and zero-shot inference mode, which directly employs fused masks without additional training.

4.3 Comparison with Different Methods

To evaluate the effectiveness of the proposed PiSOD framework, we compare it with six representative SOD models, including TRACER, InSPyReNet, M3Net, MENet, BBRF, ADMNet, 3SD, DiffSOD, DC-Net, and UniSOD. All models are tested on five public datasets: ECSSD, HKU-IS, DUT-O, PASCAL-S, and DUTS-TE. The evaluation metrics include MAE, maxF, enhanced-alignment measure (E-measure), and structure-measure (S-measure).

As shown in Table 1, PiSOD significantly improves the performance of all baseline SOD models under the fine-tuning mode. More concretely, by averaging all reported Δ improvements between PiSOD and the corresponding baseline models across different backbones and datasets, PiSOD achieves an overall absolute improvement of $\sim$0.3% in MAE and 1.3% in S-measure. These aggregated gains consistently indicate improved structural integrity and boundary preservation under diverse settings. Specifically, by introducing fusion pseudolabels generated by the SPAM module, the baseline models achieve higher accuracy and structural consistency across all metrics. This improvement is particularly evident in complex backgrounds or multiobject scenes, where the completeness of salient boundaries and semantic consistency are effectively enhanced. In the zero-shot inference mode, PiSOD can directly generate saliency maps without requiring pixel-level labels or additional training. Although its performance is slightly lower than that of the fine-tuned baseline models, its fast inference capability enables efficient operation in zero-shot or unlabeled scenarios, demonstrating the practicality of the proposed method. The visual comparisons in Fig. 4 further confirm that PiSOD produces sharper and more semantically consistent saliency maps than existing methods.

Overall, the PiSOD framework, by integrating structural and semantic priors, not only significantly enhances the performance of existing SOD models in supervised settings but also enables efficient and practical saliency prediction under unsupervised conditions. This design improves the model's generalization ability across different datasets and supervision levels, demonstrating clear advantages in terms of accuracy, robustness, and practicality. Consequently, PiSOD provides a flexible and effective solution for salient object detection tasks.

4.4 Ablation Analyses

To comprehensively evaluate the effectiveness and robustness of the proposed PiSOD framework, we conducted a series of ablation studies focusing on six key aspects: loss fusion strategy, MLLM guidance, text encoder selection, pseudolabel selection strategy, noise robustness, and detector-segmenter replacement. By systematically removing, replacing, or perturbing different modules, we quantitatively analyzed each component's contribution to overall saliency detection performance. These studies not only validate the complementary effects of semantic-structural constraints and cross-modal priors but also demonstrate the stability and adaptability of PiSOD under noisy supervision and diverse structural prior configurations.

Table 1 Quantitative comparison of PiSOD with base SOD models across multiple datasets. The best results are highlighted in bold. Rows marked with Δ indicate the performance gains of PiSOD over the corresponding baselines.

	ECSSD				HKU-IS				DUT-O				PASCAL-S				DUTS-TE			
Model	MAE ↓	F_β ↑	E_ξ ↑	S_α ↑	MAE ↓	F_β	E_ξ	S_α	MAE ↓	F_β	E_ξ	S_α	MAE ↓	F_β	E_ξ	S_α	MAE ↓	F_β	E_ξ	S_α
PiSOD (zero-shot)	0.065	0.820	0.865	0.788	0.062	0.805	0.852	0.776	0.075	0.700	0.810	0.730	0.092	0.760	0.820	0.748	0.069	0.780	0.845	0.800
$TRACER_{21}$	0.026	0.961	0.972	0.935	0.020	0.954	0.965	0.932	0.045	0.849	0.901	0.855	0.047	0.909	0.921	0.882	0.022	0.932	0.943	0.919
TRACER + PiSOD	0.024	0.965	0.985	0.946	0.020	0.964	0.976	0.942	0.042	0.856	0.911	0.877	0.048	**0.921**	0.934	0.894	**0.020**	0.943	0.956	0.928
Δ	−0.002	+0.004	+0.013	+0.011	0.000	+0.010	+0.011	+0.010	−0.003	+0.007	+0.010	+0.022	+0.001	+0.012	+0.013	+0.012	−0.002	+0.011	+0.013	+0.009
$InSPyReNet_{22}$	0.023	0.960	0.971	0.949	0.021	0.955	0.967	0.944	0.045	0.832	0.897	0.875	0.048	0.893	0.925	0.893	0.024	0.927	0.956	0.931
InSPyReNet + PiSOD	0.021	0.971	0.984	**0.957**	0.020	0.963	0.979	**0.956**	0.042	0.847	0.906	0.896	0.046	0.906	0.936	**0.903**	0.022	0.936	0.967	**0.946**
Δ	−0.002	+0.011	+0.013	+0.008	−0.001	+0.008	+0.012	+0.012	−0.003	+0.015	+0.009	+0.021	−0.002	+0.013	+0.011	+0.010	−0.002	+0.009	+0.011	+0.015
$M3Net\text{-}S_{23}$	0.021	0.947	0.974	0.948	0.019	0.937	0.977	0.943	0.045	0.811	0.903	0.872	0.047	0.864	0.932	0.889	0.024	0.902	0.960	0.927
M3Net-S + PiSOD	0.019	0.952	0.981	0.953	**0.017**	0.945	**0.982**	0.951	0.042	0.823	**0.913**	0.884	0.045	0.871	**0.942**	0.896	0.022	0.913	**0.972**	0.936
Δ	−0.002	+0.005	+0.007	+0.005	−0.002	+0.008	+0.005	+0.008	−0.003	+0.012	+0.010	+0.012	−0.002	+0.007	+0.010	+0.007	−0.002	+0.011	+0.012	+0.009
$UniSOD_{25}$	0.017	0.953	0.941	0.950	0.018	0.939	0.969	0.940	0.037	0.822	0.910	0.876	0.044	0.875	0.887	0.889	0.021	0.906	0.938	0.925
UniSOD + PiSOD	**0.016**	0.956	0.945	0.953	**0.017**	0.941	0.971	0.943	0.038	0.819	0.906	0.875	0.043	0.877	0.890	0.891	**0.020**	0.909	0.940	0.927
Δ	−0.001	+0.003	+0.004	+0.003	−0.001	+0.002	+0.002	+0.003	+0.001	−0.003	−0.004	−0.001	−0.001	+0.002	+0.003	+0.002	−0.001	+0.003	+0.002	+0.002

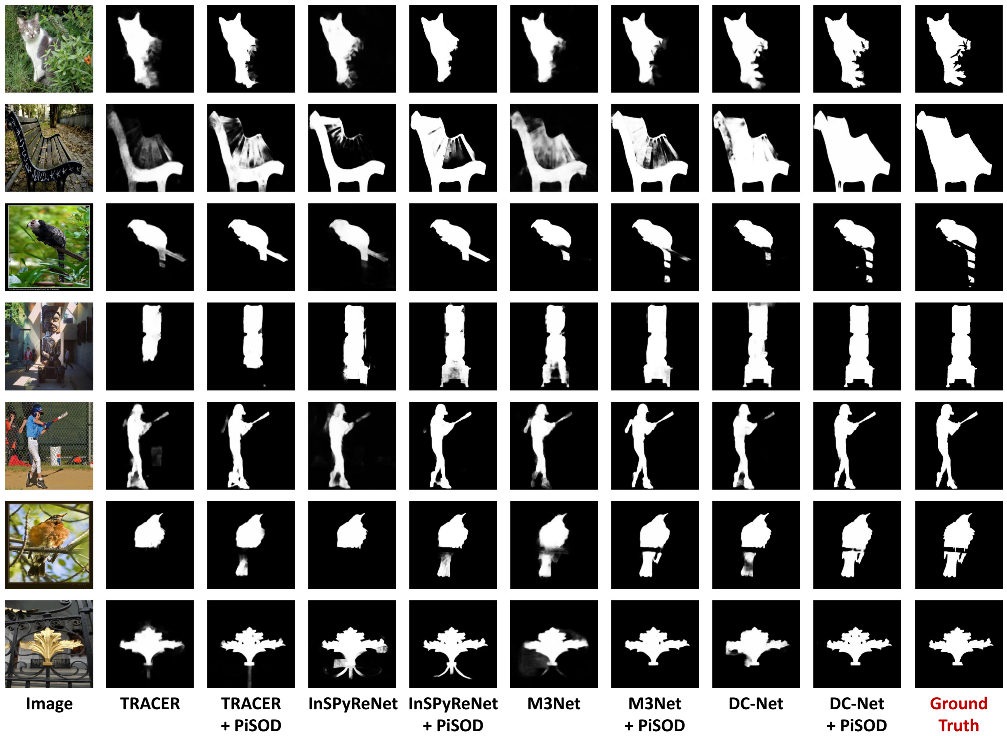

Fig. 4 Visual comparison with existing methods. Our approach produces more complete and sharper saliency maps, achieving better semantic consistency and boundary preservation.

4.4.1 *Loss fusion strategy*

We analyzed the impact of semantic and structural losses on model performance during training. To this end, we designed four training configurations: the first uses the base salient object detection model without additional losses, the second adds only the semantic loss, the third adds only the structural loss, and the fourth incorporates both semantic and structural losses. During the experiments, all other training hyperparameters were kept constant, and only the loss combinations were varied to observe their effects on saliency prediction. The results show that the full loss configuration achieves the best performance across all evaluation metrics (see Table 2). Removing the structural loss leads to reduced boundary integrity, whereas removing the semantic loss notably decreases semantic consistency in salient regions. These findings confirm that semantic and structural losses play complementary roles in enhancing overall model performance.

4.4.2 *Different Multi-Modal Language Models*

As shown in Table 3(a), to evaluate the impact of the MLLM in the semantic prior branch, we tested GPT-4, LLaVA-NeXT, Qwen-VL, Gemini 2.5, and Claude 3.5 while keeping all other modules fixed. The experiments compared the quality of pseudolabels and the final saliency prediction under different MLLM guidance, analyzing how the discriminative capability of generated text affects overall performance. The results show that GPT-4 guidance achieves the best performance in both boundary integrity and semantic consistency, whereas Gemini 2.5 and Claude 3.5 perform slightly lower than GPT-4 but still outperform LLaVA-NeXT and Qwen-VL. This demonstrates that high-quality multimodal text generation can effectively enhance the semantic accuracy of pseudolabels, thereby improving saliency detection.

4.4.3 *Different text encoders*

As shown in Table 3(b), we further thoroughly investigated the effect of text encoder selection on pseudolabel generation and saliency prediction. With the MLLM fixed as GPT-4, we employed CLIP, BLIP, Sentence-BERT, ALBEF, and E5-large to encode the textual descriptions of salient

Table 2 Comparison results show that jointly applying semantic and structural losses effectively improves saliency detection performance. All ablation experiments are conducted on the ECSSD dataset with M3Net-S as the baseline. The best results are highlighted in bold.

	Method			ECSSD				HKU-IS				DUT-O				PASCAL-S				DUTS-TE			
No.	Base	Semantic	Structural	MAE ↓	F_β ↑	E_ξ ↑	S_α ↑	MAE ↓	F_β	E_ξ	S_α	MAE ↓	F_β	E_ξ	S_α	MAE ↓	F_β	E_ξ	S_α	MAE	F_β	E_ξ	S_α
1	✓			0.021	0.947	0.974	0.948	0.019	0.937	0.977	0.943	0.045	0.811	0.903	0.872	0.047	0.864	0.932	0.889	0.024	0.902	0.960	0.927
2	✓	✓		0.020	0.949	0.976	0.950	0.019	0.936	0.976	0.943	0.044	0.814	0.905	0.874	0.047	0.864	0.934	0.886	0.023	0.903	0.962	0.928
3	✓		✓	0.020	0.945	0.973	0.945	0.020	0.935	0.971	0.938	0.045	0.805	0.896	0.870	0.049	0.866	0.933	0.891	0.024	0.901	0.961	0.931
4	✓	✓	✓	**0.019**	**0.952**	**0.981**	**0.953**	**0.017**	**0.945**	**0.982**	**0.951**	**0.042**	**0.823**	**0.913**	**0.884**	**0.045**	**0.871**	**0.942**	**0.896**	**0.022**	**0.913**	**0.972**	**0.936**

objects and then computed cosine similarity with candidate visual features to select the most relevant masks. Experimental results indicate that CLIP's text-visual joint embedding best aligns with the pseudolabel generation pipeline, yielding the most accurate saliency predictions. BLIP and Sentence-BERT achieve cross-modal matching but exhibit slightly lower boundary precision and semantic consistency, whereas ALBEF and E5-large perform moderately, yet still below CLIP. These findings highlight the importance of carefully choosing a suitable text encoder for high-quality cross-modal pseudolabels.

4.4.4 *Pseudolabel selection strategy*

As shown in Table 3(c), to assess the effectiveness of the pseudolabel selection module, we compared three candidate mask processing strategies: (1) adaptive maximum adjacent difference, (2) fixed threshold, and (3) no selection (direct fusion of all candidate masks). The results show that the adaptive strategy automatically selects the most salient masks based on semantic-structural alignment, improving pseudolabel quality and final saliency prediction performance. The fixed-threshold strategy performs reasonably on some datasets but is generally less robust than the adaptive approach. Without any selection, noise is introduced, leading to a noticeable drop in saliency prediction performance.

Table 3 Panels (a), (b), and (c) show performance variations under different multi-modal language models, text encoders, and pseudolabel selection strategies. Results show that model and encoder choices significantly affect overall performance, whereas adaptive pseudolabel filtering further improves detection accuracy. All ablation experiments are conducted on the ECSSD dataset with M3Net-S as the baseline. The best results are highlighted in bold.

a. Effect of different MLLMs			
MLLM	$F_\beta \uparrow$	$E_\xi \uparrow$	$S_\alpha \uparrow$
Qwen-VL	90.2	94.1	90.9
LLaVA-NeXT	91.8	95.4	92.5
Claude 3.5	93.1	96.8	93.7
Gemini 2.5	94.3	97.5	94.6
GPT-4o	**95.2**	**98.1**	**95.3**
b. Effect of different text encoders			
Text encoder	$F_\beta \uparrow$	$E_\xi \uparrow$	$S_\alpha \uparrow$
E5-large	93.8	96.7	94.1
ALBEF	94.4	96.8	94.3
Sentence-BERT	93.1	96.2	93.5
BLIP	94.8	97.6	94.9
CLIP	**95.2**	**98.1**	**95.3**
c. Effect of pseudolabel selection strategies			
Selection	$F_\beta \uparrow$	$E_\xi \uparrow$	$S_\alpha \uparrow$
No filtering	92.8	95.5	93.3
Fixed threshold	94.1	96.9	94.4
AMAD	**95.2**	**98.1**	**95.3**

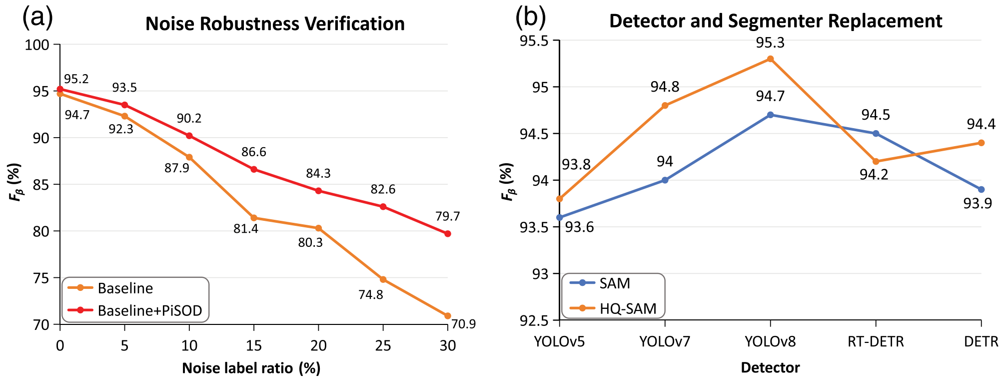

Fig. 5 (a) and (b) Performance variations under different noise levels and detector segmenter combinations, highlighting PiSOD's robustness and the impact of structure prior choices on saliency detection. All ablation experiments are conducted on the ECSSD dataset with M3Net-S as the baseline. 3

4.4.5 *Noise robustness verification*

As shown in Fig. 5(a), to evaluate the robustness of PiSOD under noisy supervision, we introduced pixel-level random label noise into the DUTS-TR training set, ranging from 0% to 30% in increments of 5%, while keeping the testing set clean. Both the baseline and baseline+PiSOD models were fine-tuned under identical settings, and their F_β scores were compared to assess performance degradation under different noise levels. The results show that the baseline model suffers a sharp decline in accuracy and boundary precision as noise increases, whereas the model equipped with PiSOD maintains much higher stability and semantic consistency. This indicates that PiSOD effectively suppresses the negative effects of noisy supervision, enhancing the robustness of salient object detection under imperfect annotations.

4.4.6 *Detector and segmenter replacement*

As shown in Fig. 5(b), to investigate the influence of different structural prior configurations, we replaced the detection and segmentation components in the structural prior branch. Specifically, we tested several detectors, including YOLOv5, YOLOv7, YOLOv8, RT-DETR, and DETR, each combined with two segmentation models: SAM and HQ-SAM. The experimental results reveal that both the detection accuracy and segmentation granularity substantially affect the quality of generated structural priors. The combination of YOLOv8 and HQ-SAM achieves the highest F_β score of 95.3%, owing to its precise object localization and high-fidelity mask generation. By contrast, configurations using earlier detectors (e.g., YOLOv5, YOLOv7) or the basic SAM model tend to yield less detailed boundaries and lower structural completeness. These observations indicate that stronger detectors coupled with advanced segmentation models can better capture object structures, thereby enhancing the robustness and consistency of PiSOD in pseudolabel generation.

From the above ablation studies, it can be concluded that each component of the PiSOD framework plays a vital role in enhancing model accuracy, stability, and generalization. The loss fusion strategy, MLLM guidance, text encoder selection, and adaptive pseudolabel selection collectively ensure semantic coherence and structural integrity, whereas the noise robustness and detector–segmenter replacement experiments further demonstrate the framework's resilience to label imperfections and its flexibility across different structural priors. Removing or altering any of these modules leads to a noticeable decline in performance, confirming the effectiveness and complementarity of all core designs in PiSOD.

4.5 Computational Cost Analysis

To analyze the computational efficiency of PiSOD, we report the time overhead of the additional modules and backbone-related stages in Table 4. As shown in Table 4(a), although PiSOD introduces several auxiliary components, including MLLM guidance, object detection, SAM-based segmentation, and pseudolabel fusion, none of these modules are involved during inference, as they are only utilized for offline pseudolabel generation.

Table 4 Panels (a) and (b) summarize the computational overhead of PiSOD, where (a) reports the time cost of the additional modules and (b) presents the backbone-related computational cost across different stages.

a. Computational overhead of additional modules in PiSOD

Category	Module	Time (ms/image)	Used for inference
Semantic prior	MLLM guidance (GPT4o)	114.58752	No
Structural prior	Object detector (YOLO)	48.67572	No
	SAM segmentation	41.61761	No
Pseudolabel fusion	Alignment module (SPAM)	53.02569	No

b. Backbone-related computational cost across different stages

Stage	Description	Time (ms/image)	Affected by PiSOD
Backbone pretraining	Original SOD training	214.17039	No
Pseudolabel generation	PiSOD pseudolabeling	223.35418	Yes
Fine-tuning	Backbone fine-tuning	306.49635	Yes
Inference	SOD model inference	76.02724	No

Table 4(b) further presents the backbone-related computational cost across different stages. Compared with standard SOD training, PiSOD introduces additional overhead during pseudolabel generation and backbone fine-tuning, which is expected due to the incorporation of semantic and structural priors. However, the inference time remains almost unchanged, demonstrating that PiSOD does not introduce an extra runtime burden during deployment.

Overall, PiSOD achieves significant performance gains with negligible inference overhead, making it practical for real-world saliency detection applications.

5 Conclusion

In this work, we proposed a novel pseudosupervised and easily integrable salient object detection framework, PiSOD, along with a modular Saliency Prior Alignment Module (SPAM). By integrating object-level structural and semantic priors, PiSOD generates semantically consistent and boundary-preserving saliency maps, significantly enhancing performance when fine-tuning existing SOD models and enabling reliable zero-shot saliency predictions. Extensive experiments across five benchmark datasets demonstrate that, when averaged over different backbone networks and datasets, PiSOD consistently improves the baseline models by ~0.3% in MAE and 1.3% in S-measure, reflecting clearer object boundaries and better structural integrity. In addition, ablation studies confirm the effectiveness of each key module in contributing to these gains. Overall, the PiSOD framework provides an efficient, flexible, and scalable solution for saliency detection under weakly supervised or label-free conditions. Future work may explore improved cross-modal fusion, video-based saliency detection, and lightweight deployment strategies to further enhance practical applicability and overall performance.

Disclosures

The authors have no conflicts of interest to declare that are relevant to the content of this article.

Code and Data Availability

The code, data, and materials will be provided upon request or can be obtained directly from the corresponding author.

References

1. L. Itti, C. Koch, and E. Niebur, "A model of saliency-based visual attention for rapid scene analysis," *IEEE Trans. Pattern Anal. Mach. Intell.* **20**(11), 1254–1259 (1998).
2. A. Borji et al., "Salient object detection: a survey," *Comput. Vis. Media* **5**(2), 117–150 (2019).
3. M.-M. Cheng et al., "Global contrast based salient region detection," *IEEE Trans. Pattern Anal. Mach. Intell.* **37**(3), 569–582 (2015).
4. L. Wang et al., "Learning to detect salient objects with image-level supervision," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, pp. 136–145 (2017).
5. D.-P. Fan et al., "Re-thinking co-salient object detection," *IEEE Trans. Pattern Anal. Mach. Intell.* **44**(8), 4339–4354 (2022).
6. G. Li and Y. Yu, "Deep contrast learning for salient object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, pp. 478–487 (2016).
7. Z. Liu et al., "Swin Transformer: hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, pp. 10012–10022 (2021).
8. J. Zhang et al., "UC-Net: uncertainty inspired RGB-D saliency detection via conditional variational autoencoders," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, pp. 8582–8591 (2020).
9. J. Zhang et al., "Weakly-supervised salient object detection via scribble annotations," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, pp. 12546–12555 (2020).
10. J.-J. Liu et al., "A simple pooling-based design for real-time salient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, pp. 3917–3926 (2019).
11. X. Qin et al., "U2-Net: going deeper with nested U-structure for salient object detection," *Pattern Recognit.* **106**, 107404 (2020).
12. B. Xu et al., "Locate globally, segment locally: a progressive architecture with knowledge review network for salient object detection," *Proc. AAAI Conf. Artif. Intell.* **35**(4), 3004–3012 (2021).
13. A. Kirillov et al., "Segment anything," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, pp. 4015–4026 (2023).
14. L. Ke et al., "Segment anything in high quality," in *Adv. Neural Inf. Process. Syst. 36*, pp. 29914–29934 (2023).
15. Q. Hou et al., "Deeply supervised salient object detection with short connections," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, pp. 3203–3212 (2017).
16. Z. Chen et al., "Global context-aware progressive aggregation network for salient object detection," *Proc. AAAI Conf. Artif. Intell.* **34**(07), 10599–10606 (2020).
17. P. Zhang et al., "Amulet: aggregating multi-level convolutional features for salient object detection," in *Proc. IEEE Int. Conf. Comput. Vis.*, pp. 202–211 (2017).
18. Y. Pang et al., "Multi-scale interactive network for salient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, pp. 9413–9422 (2020).
19. T. Wang et al., "Detect globally, refine locally: a novel approach to saliency detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, pp. 3127–3135 (2018).
20. J.-X. Zhao et al., "EGNet: edge guidance network for salient object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, pp. 8779–8788 (2019).
21. O. Ronneberger, P. Fischer, and T. Brox, "U-Net: convolutional networks for biomedical image segmentation," *Lect. Notes Comput. Sci.* **9351**, 234–241 (2015).
22. N. Liu, J. Han, and M.-H. Yang, "PiCANet: learning pixel-wise contextual attention for saliency detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, pp. 3089–3098 (2018).
23. X. Qin et al., "BASNet: boundary-aware salient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, pp. 7479–7489 (2019).
24. Q. Yan et al., "Hierarchical saliency detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, pp. 1155–1162 (2013).
25. Y. Li et al., "The secrets of salient object segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, pp. 280–287 (2014).
26. L. Zhang et al., "A bi-directional message passing model for salient object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, pp. 1741–1750 (2018).
27. X. Zhang et al., "Progressive attention guided recurrent network for salient object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, pp. 714–722 (2018).
28. S. Yu et al., "Structure-consistent weakly supervised salient object detection with local saliency coherence," *Proc. AAAI Conf. Artif. Intell.* **35**(4), 3234–3242 (2021).
29. Y. Zeng et al., "Multi-source weak supervision for saliency detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, pp. 6074–6083 (2019).
30. C. Yang et al., "Saliency detection via graph-based manifold ranking," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, pp. 3166–3173 (2013).
31. J. Wei, S. Wang, and Q. Huang, "Net: fusion, feedback and focus for salient object detection," *Proc. AAAI Conf. Artif. Intell.* **34**(07), 12321–12328 (2020).

32. Y. Wang et al., "Self-supervised equivariant attention mechanism for weakly supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, pp. 12275–12284 (2020).
33. Z. Zhang et al., "Bilateral attention network for RGB-D salient object detection," *IEEE Trans. Image Process.* **30**, 1949–1961 (2021).
34. J. Redmon et al., "You only look once: unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, pp. 779–788 (2016).
35. J. Redmon and A. Farhadi, "YOLOv3: an incremental improvement," arXiv:1804.02767 (2018). 5
36. A. Radford et al., "Learning transferable visual models from natural language supervision," in *Int. Conf. Mach. Learn.*, PMLR, pp. 8748–8763 (2021).
37. J. Li et al., "BLIP: bootstrapping language-image pre-training for unified vision-language understanding and generation," in *Int. Conf. Mach. Learn.*, PMLR, pp. 12888–12900 (2022).
38. S. Doveh et al., "Towards multimodal in-context learning for vision and language models," *Lect. Notes Comput. Sci.* **15641**, 250–267 (2024).
39. C. Jia et al., "Scaling up visual and vision-language representation learning with noisy text supervision," in *Int. Conf. Mach. Learn.*, PMLR, pp. 4904–4916 (2021).
40. T. Brown et al., "Language models are few-shot learners," in *Adv. Neural Inf. Process. Syst. 33*, pp. 1877–1901 (2020).
41. G. Li and Y. Yu, "Visual saliency based on multiscale deep features," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, pp. 5455–5463 (2015).
42. F. Perazzi et al., "Saliency filters: contrast based filtering for prominent region detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, pp. 733–740 (2012).
43. R. Achanta et al., "Frequency-tuned salient region detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, pp. 1597–1604 (2009).
44. D.-P. Fan et al., "Enhanced-alignment measure for binary foreground map evaluation," arXiv:1805.10421 (2018).
45. D.-P. Fan et al., "Structure-measure: a new way to evaluate foreground maps," in *Proc. IEEE Int. Conf. Comput. Vis.*, pp. 4548–4557 (2017).
46. C. Shorten and T. M. Khoshgoftaar, "A survey on image data augmentation for deep learning," *J. Big Data* **6**(1), 60 (2019).
47. M. S. Lee, W. Shin, and S. W. Han, "Tracer: extreme attention guided salient object tracing network (student abstract)," *Proc. AAAI Conf. Artif. Intell.* **36**(11), 12993–12994 (2022).
48. T. Kim et al., "Revisiting image pyramid structure for high resolution salient object detection," in *Proc. Asian Conf. Comput. Vis.*, pp. 108–124 (2022).
49. Y. Yuan, P. Gao, and X. Tan, "Net: multilevel, mixed and multistage attention network for salient object detection," arXiv:2309.08365 (2023).
50. Y. Wang et al., "Pixels, regions, and objects: multiple enhancement for salient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, pp. 10031–10040 (2023).
51. M. Ma et al., "Boosting broader receptive fields for salient object detection," *IEEE Trans. Image Process.* **32**, 1026–1038 (2023).
52. X. Zhou, K. Shen, and Z. Liu, "ADMNet: attention-guided densely multi-scale network for lightweight salient object detection," *IEEE Trans. Multimed.* **26**, 10828–10841 (2024).
53. R. Yasarla et al., "3SD: self-supervised saliency detection with no labels," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis.*, pp. 313–322 (2024).
54. M. Song et al., "Pushing the boundaries of salient object detection: a denoising-driven approach," *IEEE Trans. Image Process.* **34**, 3903–3917 (2025).
55. J. Zhu, X. Qin, and A. Elsaddik, "DC-Net: divide-and-conquer for salient object detection," *Pattern Recognit.* **157**, 110903 (2025).
56. K. Wang et al., "Unified-modal salient object detection via adaptive prompt learning," *IEEE Trans. Circuits Syst. Video Technol.* **35**, 12109–12124 (2025).

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